Number System Explorer

How Have Number Systems Changed Over Time?

Name:	Grade/Section:	Date:

Inquiry Statement :

How have number systems changed over time, and why did different civilizations create their own ways to represent numbers?

Global Context

Orientation in Space and Time

Assessment Focus

Criterion A – Knowing & Understanding | Criterion B – Investigating Patterns

Your Task

Throughout history, different civilizations developed their own number systems to solve everyday problems such as counting, trading, measuring land, building structures, and recording information. From Roman numerals to the Hindu-Arabic number system we use today, numbers have evolved to make life easier.

Your challenge is to become a Number System Explorer and investigate how mathematics has developed over time while applying different number concepts to solve real-life problems.

Step 1 —Choose ONE ancient civilization.

☐ Egyptians ☐ Romans ☐ Babylonians	☐ Mayans ☐ Ancient Indians ☐Chinese

Other (teacher-approved):

Section 1 — Journey Through Number Systems (Research)

Research and answer the following in 40–50 words.

Which civilization did you choose?	
What number system did they use?	
Why was it important?	
One interesting fact	

Word count check: aim for 10–20 words for each of your answers above.

Section 2 – Explore the Mathematics

Your innovation must include at least SIX of the mathematical concepts below. Tick the six (or more) you will use, then complete the table with a real example from your innovation.

☐ Whole Numbers ☐ Natural Numbers ☐ Integers ☐ Fractions, Decimals ☐ Percentages	☐ Profit and Loss ☐ Number Properties ☐ Factors and Multiples ☐ Prime numbers, prime Factorisation ☐ LCM ☐ HCF

Mathematics Concept	Example from Your Innovation

Section 3 – Number Investigation

Choose five numbers from your research or create your own examples

Number	Number Type (Natural / Whole / Integer / Fraction / Decimal)	Why did you choose it?)

Section 4 – Mathematical Investigation

Choose TWO numbers from your Section 3 research and investigate them mathematically.

Investigation A – Factors and Multiples

My two numbers	
Factors	
Multiples	
HCF	
LCM	
What pattern did you notice?	

Investigation B – Prime numbers

Choose one composite number	
Prime Factorisation	
It is a prime or composite	
My observation	

Section 5 –Fractions, Decimals, Percentage and Profit & Loss

Fraction Investigation

Choose one fraction	
Convert into decimal and percentage	
Where might you use it in real life?	

Profit & Loss Investigation

Buying price	
Selling price	
Profit/ Loss	
Explain	

Section 6 – Discover the patterns

Complete the table

Property	Example	What did you notice?
Commutative		
Associative		
Identity		
Distributive		

Section 7 – Think Like a Mathematician

Complete the reflection table below.

I discovered...	**Evidence**
I discovered that number systems have changed because...	
The most useful mathematical concept was...	
One pattern I noticed was...	
Mathematics helps people because...	

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Assessment Rubric (Formative)

Criteria	Emerging (1)	Developing (2)	Proficient (3)	Excellent (4)
Research Skills	Limited information; little relevance.	Basic information from one source.	Relevant information from multiple sources.	Thorough, accurate research with interesting insights.
Mathematical Understanding (Criterion A)	Demonstrates limited understanding of concepts	Demonstrates some understanding with minor errors.	Correctly applies most mathematical concepts	Accurately explains and applies mathematical concepts confidently.
Pattern Investigation (Criterion B)	Limited observations and conclusions.	Identifies simple patterns.	Explains patterns with evidence.	Makes clear generalisations and justifies conclusions using evidence.
Connection to orientation in Space and Time	Limited connection to historical development of number systems.	Some mathematical links are made.	Clear explanation of how the number system evolved.	Insightful explanation showing how historical developments influenced modern mathematics.
Presentation & Communication	Difficult to follow; incomplete.	Adequately organised.	Well-organised, neat and easy to understand.	Highly engaging, creative and logically presented.